# Group I, Row V, part 2: check of 2a, 2b (UV) and 2b (finite, plus prescription)

Notation:

- $L \equiv \log \frac{\vee}{\Lambda} + \log \frac{k^+-\Lambda}{\vee} = \log \frac{k^+-\Lambda}{\Lambda} = \int_\Lambda^{k^+-\Lambda} \frac{dp^+}{p^+}$, and $\lambda \equiv \frac{\Lambda}{k^+}$;
- $D \equiv p^+(y-x)^2 + (k^+-p^+)(y-z)^2$, and $D_q \equiv (k^+-q^+)(y-x)^2 + q^+(y-z)^2$ with $q^+ = k^+ - p^+$;
- $B_1, \ldots, B_6$ are the six brackets of your expression, in the order written. $B_2 = \left[\frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]$ carries $-\frac{p^+}{k^+-p^+}$, and $B_4 = \left[\frac{(y-z)^i(x-z)^m}{(x-z)^2} + \frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]$ carries $-\frac{k^+-p^+}{p^+}$;
- the colour operators are

$$\mathcal{O}_x \equiv \left[U^{ab'}(x') - U^{ab'}(w')\right]\left[N_c U^{af}(y) - N_c U^{af}(x)\right]\rho^{b'}(x')\rho^f(y),$$
$$\mathcal{O}_z \equiv \left[U^{ab'}(x') - U^{ab'}(w')\right]\left[N_c U^{af}(y) - N_c U^{af}(z)\right]\rho^{b'}(x')\rho^f(y).$$

Corrections are in blue; removed text is in red.

## 1 Verdict, step by step

| step | result |
|---|---|
| identity $f^{abc}f^{def}U^{be}(z)U^{cf}(z) = N_c U^{ad}(z)$ | correct. It is the invariance $f^{abc}U^{bb'}U^{cc'} = U^{aa'}f^{a'b'c'}$ plus $f^{gef}f^{def} = N_c\delta^{gd}$. At $x = z$, $-f^{adc}f^{fbe}U^{db}U^{ce} = -N_c U^{af}(z)$. |
| split into 2a ($+N_cU(x)$) and 2b ($-N_cU(x)$) | correct. The 2a colour factor vanishes at $x \to z$, so 2a is UV finite; 2b keeps the divergence. |
| 2a with the plus prescription | **correct** up to $O(\Lambda/k^+)$ from the limits $0, k^+$ (rel. $9.8 \times 10^{-4}$ at $\Lambda/k^+ = 10^{-3}$, $1.3 \times 10^{-5}$ at $10^{-5}$). Signs, phases ($e^{-ik\cdot(w'-z)}$ at $p^+ \to 0$, $e^{-ik\cdot(w'-x)}$ at $p^+ \to k^+$), $\frac{1}{(y-z)^2}$, $\frac{1}{(y-x)^2}$ and the prefactor $\frac{g^4}{16\pi^5}$ are all right. |
| 2b UV term ($d_\perp = 2-\epsilon$) | correct (`Part2b_MSbar.pdf`, `Part2b_UV_finite.pdf`). |
| 2b UV, "setting $\Lambda \to 0$" | inconsistent (M1). |
| 2b finite before "Simplifying" | correct, with $d_\perp = 2$ in the finite term (M4). |
| 2b finite, "Simplifying" | **error E1**: the $B_2$ pieces get $\mathcal{O}_z$; they must keep $\mathcal{O}_x$. |
| 2b finite, final | **errors E1, E2, E3**. Your form is off by 0.2–3.7 (relative). With all three fixes it agrees with the exact $p^+$ integral to $1.6\times10^{-6}$ at $\Lambda/k^+ = 10^{-5}$; the residue is the $O(\Lambda/k^+)$ of your limits. The corrected form below with exact limits agrees to $3 \times 10^{-13}$. |

## 2 The errors

**E1. Colour of the $B_2$ pieces (2b finite).**

- The $B_2$ term is $-\frac{p^+}{k^+-p^+}B_2 = -\frac{k^+}{q^+}B_2 + B_2$, with $q^+ = k^+ - p^+$.
- Its colour factor is that of 2b, $[N_cU(y) - N_cU(x)]$, i.e. $\mathcal{O}_x$. The colour does not depend on $p^+$.
- In your "Simplifying" step and in the final form, the three $B_2$ pieces have $[N_cU(y) - N_cU(z)]$:
  - the $[\frac{k^+}{q^+}]_+$ term;
  - the $-B_2$ term;
  - the first blue endpoint term.

  All three must have $[N_cU(y) - N_cU(x)]$. Only the counterterm has $\mathcal{O}_z$.

**E2. The $p^+ \to 0$ endpoint of $B_4$ (2b finite).** At $p^+ = 0$:

- the phase is $e^{-ik\cdot(w'-z)}e^{-ik\cdot(z-x)\cdot 0} = e^{-ik\cdot(w'-z)}$;
- $D = k^+(y-z)^2$;
- the colour is $\mathcal{O}_x$.

Your second blue term has the right kernel and $\frac{1}{(y-z)^2}$, but $e^{-ik\cdot(w'-x)}$ and $\mathcal{O}_z$. The correct factors are $e^{-ik\cdot(w'-z)}$ and $\mathcal{O}_x$. (Your 2a has it right: $e^{-ik\cdot(w'-z)}$ for the $p^+ \to 0$ term and $e^{-ik\cdot(w'-x)}$ for the $p^+ \to k^+$ term.)

**E3. The counterterm after the $p^+$ integral.** The counterterm depends on $p^+$ only through $\frac{1}{k^+}\big[\frac{p^+(k^+-p^+)}{k^{+2}} + \frac{p^+}{k^+-p^+} + \frac{k^+-p^+}{p^+}\big]$. Exactly,

$$\int_\Lambda^{k^+-\Lambda} \frac{dp^+}{k^+}\Big[\frac{p^+(k^+-p^+)}{k^{+2}} + \frac{p^+}{k^+-p^+} + \frac{k^+-p^+}{p^+}\Big] = 2L - \frac{11}{6} + 4\lambda - \lambda^2 + \frac{2}{3}\lambda^3. \tag{2.1}$$

So the last term of your final form must be

$$-\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\Big[\frac{11}{6} - \frac{1}{3}L\ 2L - \frac{11}{6}\Big]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}\mathcal{O}_z,$$

plus $4\lambda - \lambda^2 + \frac{2}{3}\lambda^3$ inside the bracket if $\Lambda$ is kept. In terms of where the pieces come from:

- the $\frac{1}{p^+}$ and $\frac{1}{k^+-p^+}$ parts give $L$ each, i.e. $2L$; your plus prescription on them gives zero, because nothing else in the counterterm depends on $p^+$;
- the rest gives $\int_0^1 d\xi[\xi(1-\xi)-2] = -\frac{11}{6}$.

**Minor points.**

M1. In "Setting $\Lambda \to 0$" (2b UV) you drop $\frac{4\Lambda}{k^+} - \frac{\Lambda^2}{k^{+2}} + \frac{2\Lambda^3}{3k^{+3}}$ but keep $2\log(1-\frac{\Lambda}{k^+})$, which is of the same order. Drop both, or keep both.

M2. The plus prescription is $\int dp^+[\frac{k^+}{p^+}]_+ f \equiv \int dp^+ \frac{k^+}{p^+}[f(p^+) - f(0)]$, with the factor $k^+$. Here $f$ is everything else that depends on $p^+$: the phase, $\frac{1}{D}$ and the colour-independent brackets. The same holds for $[\frac{k^+}{q^+}]_+$ at $q^+ = 0$.

M3. The phase $e^{-ik\cdot(z-x)p^+/k^+}$ depends on $x$, $z$ and $p^+$. It belongs inside $\int dp^+ \int_{x,z}$, not in front of them.

M4. If $d_\perp$ is kept general in $B_1$, the counterterm has $\frac{d_\perp}{2}\frac{p^+(k^+-p^+)}{k^{+2}}$ (`Part2b_UV_finite.pdf`). With $d_\perp = 2$ in the finite term, your form is right.

M5. **Finiteness piece by piece.**

- *UV.* With the counterterm integrated over $p^+$ as a separate term, the finite part is UV finite only as a sum. The $B_4$ and $B_2$ endpoint terms ($\mathcal{O}_x$), the integrated counterterm ($\mathcal{O}_z$) and the regular terms each behave like $\frac{1}{(x-z)^2}$ at $x \to z$.
- *Large distances.* The endpoint terms fall only like $\frac{1}{2(x-z)^2}$; the plus-prescribed part has the opposite behaviour.
- The grouping of Sec. 4 avoids the UV problem: every piece there is UV finite (`PlusPrescription_2b.pdf`).

# 3 2b finite, corrected (your structure)

$$\begin{aligned}
\frac{d^3N_{2b\,\text{finite}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\frac{1}{D}\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
\times\Bigg[\frac{\delta^{im}d_\perp}{2(x-z)^2}\big[(y-x)^2-(y-z)^2\big]\frac{p^+(k^+-p^+)}{k^{+2}}\\
+\frac{(k^+-p^+)\delta_{im}}{k^+}\Big[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\Big]\\
-\Big[\frac{k^+}{p^+}\Big]_+\Big[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\Big]+\Big[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\Big]\\
+\frac{p^+\delta_{im}}{k^+}\Big[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\Big]
\end{aligned}$$

$$
\begin{aligned}
&- \left[\frac{(y-x)^m(x-z)^i}{(x-z)^2} - \frac{(y-x)^m(y-z)^i}{2(y-z)^2} + \frac{(y-z)^m(x-z)^i}{(x-z)^2} + \frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\mathcal{O}_x \\
&+ \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_{k^+}^{0}\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}e^{-ik\cdot(z-x)\frac{k^+-q^+}{k^+}}\frac{1}{D_q}\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2} \\
&\quad\times\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\left[\frac{k^+}{q^+}\right]_+\mathcal{O}_x \\
&- \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_{k^+-\Lambda}^{\Lambda}\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}e^{-ik\cdot(z-x)\frac{k^+-q^+}{k^+}}\frac{1}{D_q}\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2} \\
&\quad\times\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\mathcal{O}_x \\
&- \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}L\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\frac{1}{(y-x)^2}\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\mathcal{O}_x \\
&- \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}L\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\frac{1}{(y-z)^2}\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2} + \frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]\mathcal{O}_x \\
&- \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\left[2L - \frac{11}{6}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)} \\
&\quad\times\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}\,\mathcal{O}_z. \qquad (3.1)
\end{aligned}
$$

- With $\Lambda$ kept: the second line runs over $\Lambda < q^+ < k^+ - \Lambda$, and the last bracket gets $+4\lambda - \lambda^2 + \frac{2}{3}\lambda^3$. This exact form agrees with your 2b finite before the plus prescription to $3\times10^{-13}$ (Sec. 6).
- The two endpoint kernels simplify:
  - $\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2} + \frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right] = \frac{(x'-w')\cdot(y-z)}{(x'-w')^2}\left[-\frac{1}{(x-z)^2} + \frac{(y-x)\cdot(z-x)}{2(y-x)^2(z-x)^2}\right]$;
  - $\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right] = \frac{(x'-w')\cdot(y-x)}{(x'-w')^2}\left[-\frac{1}{(x-z)^2} + \frac{(y-z)\cdot(x-z)}{2(y-z)^2(x-z)^2}\right]$.

## 4 Recommended grouping (each piece UV finite)

Split the counterterm's $2L$ into its two endpoints ($L$ each), keep its regular part $\frac{p^+(k^+-p^+)}{k^{+2}} - 2$ under the $p^+$ integral, and put the endpoints together. With $K(a,b,c) \equiv \frac{(a-c)\cdot(b-c)}{(a-c)^2(b-c)^2}$, the endpoint terms become

$$
\begin{aligned}
&\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}L\int_{x,y,z,x',w'}\Bigg\{e^{-ik\cdot(w'-z)}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\Big[\big(K(z,z,x) - \tfrac{1}{2}K(y,z,x)\big)\mathcal{O}_x - \big(K(z,z,x) - K(y,z,x)\big)\mathcal{O}_z\Big] \\
&\quad + e^{-ik\cdot(w'-x)}\frac{(x'-w')\cdot(y-x)}{(x'-w')^2(y-x)^2}\big(K(x,x,z) - \tfrac{1}{2}K(y,x,z)\big)\mathcal{O}_x \\
&\quad - e^{-ik\cdot(w'-z)}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\big(K(z,z,x) - K(y,z,x)\big)\mathcal{O}_z\Bigg\}. \qquad (4.1)
\end{aligned}
$$

- In each line, the $\mathcal{O}_x$ and $\mathcal{O}_z$ terms cancel at $x \to z$, so each line is UV finite.
- The plus-prescribed integral, with the counterterm's regular part $-\frac{1}{k^+}\big[\frac{p^+(k^+-p^+)}{k^{+2}} - 2\big](\dots)\mathcal{O}_z$ kept inside it, is also UV finite.
- This is the endpoint result of `PlusPrescription_2b.pdf`, Sec. 4.2, with $\ln\frac{k^+-\Lambda}{\Lambda} = L$.

## 5 The $\log\frac{\vee}{\Lambda}$ terms (evolution)

With $L = \log\frac{\vee}{\Lambda} - \log\frac{\vee}{k^+} + \log(1-\frac{\Lambda}{k^+})$, the $\log\frac{\vee}{\Lambda}$ terms of part 2 are the following.

- **2a:** its two endpoint terms with $L \to \log\frac{\vee}{\Lambda}$. Your expression is correct; its colour is $\big[U^{ab'}(x') - U^{ab'}(w')\big]\big[-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z) + N_c U^{af}(x)\big]\rho^{b'}(x')\rho^f(y)$.
- **2b finite:** Eq. (4.1) (or the last three lines of Eq. (3.1)) with $L \to \log\frac{\vee}{\Lambda}$. The counterterm gives $2\log\frac{\vee}{\Lambda}$.
- **2b UV:** $\frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\,2\log\frac{\vee}{\Lambda}\int e^{-ik\cdot(w'-z)}\big[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu^2}{4}\big]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}[U(x') - U(w')][U(y) - U(z)]\rho\rho$.

- **The pole of the last term** is not a renormalization; it is not removed by $\overline{\text{MS}}$. In $d$ dimensions it is exactly the $x$ integral of the counterterm's $2\log\frac{\vee}{\Lambda}$, which appears with the opposite sign in the 2b finite term. So the $\log\frac{\vee}{\Lambda}$ content of 2b (UV + finite) is that of your original 2b: the two endpoint terms with $\mathcal{O}_x$. These $\frac{1}{(x-z)^2}$ singularities cancel only in the sum of all $\log\frac{\vee}{\Lambda}$ two-$\rho$ terms (JIMWLK$\otimes$LO), as discussed in `Part2b_MSbar.pdf`, Sec. 7.4.

# 6 Checks

Script `part2_user_check.py` → `part2_user_check.json`.

- Setup: 6 random sets of transverse points $x, y, z, x', w'$, random $\mathbf{k}$, $k^+ = 1.7$, $d_\perp = 2$.
- The colour operators are independent random complex numbers $\mathcal{O}_a$, $\mathcal{O}_x$, $\mathcal{O}_z$, since they do not depend on $p^+$.
- Exact reference: your 2a and your 2b finite (before the plus prescription), integrated over $\Lambda < p^+ < k^+ - \Lambda$.

| max. relative deviation from the exact $p^+$ integral | $\Lambda/k^+ = 0.05$ | $10^{-3}$ | $10^{-5}$ |
|---|---|---|---|
| your 2a (plus prescription) | $2.1 \times 10^{-1}$ | $9.8 \times 10^{-4}$ | $1.3 \times 10^{-5}$ |
| 2b finite, corrected, exact limits (Eq. (3.1)) | $8.7 \times 10^{-16}$ | $4.3 \times 10^{-15}$ | $3.4 \times 10^{-13}$ |
| your 2b finite, final form | $3.7 \times 10^{0}$ | $1.9 \times 10^{0}$ | $1.9 \times 10^{0}$ |
| with E1 fixed | $4.8 \times 10^{0}$ | $1.9 \times 10^{0}$ | $1.5 \times 10^{0}$ |
| with E1, E2 fixed | $7.6 \times 10^{-1}$ | $9.8 \times 10^{-1}$ | $1.1 \times 10^{0}$ |
| with E1, E2, E3 fixed | $7.9 \times 10^{-2}$ | $8.6 \times 10^{-5}$ | $1.6 \times 10^{-6}$ |

Your 2a and the fully fixed 2b converge to the exact result like $\Lambda/k^+$. The corrected 2b with exact limits is exact for any $\Lambda$.